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Course outline

How does an NPTEL online course work?

Week 0

Week 1

- Basic Concepts of Integrated Circuit - I
- Basic Concepts of Integrated Circuit - II
- Overview of VLSI Design Flow - I
- Overview of VLSI Design Flow - II
- Tutorial 1
- Quiz: Week 1: Assignment 1
- Lecture Materials
- Week 1 Feedback Form : VLSI Design Flow: RTL to GDS
- Week 1: Solution

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Week 1: Assignment 1

The due date for submitting this assignment has passed.

Due on 2023-08-09, 23:59 IST.

As per our records you have not submitted this assignment.

1) Which of the following statements **best describes** Moore's prediction or Moore's law: *1 point*

- Every action has an equal and opposite reaction.
- The number of components in an IC can never be increased.
- The number of components in an IC realized at minimum cost will double every year or every two years.
- The sizes of transistors in ICs realized at minimum cost will remain constant.

No, the answer is incorrect.
Score: 0
Accepted Answers:
The number of components in an IC realized at minimum cost will double every year or every two years.

2) What is the purpose of masks in photolithography: *1 point*

- Define shapes/patterns that need to be obtained on the wafer.
- Isolate multiple layers of interconnect.
- Etch silicon during chemical etching.
- Oxidize silicon after photolithography.

No, the answer is incorrect.
Score: 0
Accepted Answers:
Define shapes/patterns that need to be obtained on the wafer.

3) Which of the following statements is **incorrect**: *1 point*

- Multiple dies are fabricated on a single silicon wafer.
- Silicon wafers are sliced out from silicon ingots.
- We get silicon ingots from a silicon die.
- We obtain a chip after packaging a die.

No, the answer is incorrect.
Score: 0
Accepted Answers:
We get silicon ingots from a silicon die.

4) Which of the following statements is **incorrect**: *1 point*

- The semiconductor manufacturing plant where integrated circuits are fabricated is called a semiconductor foundry.
- Fabless design companies design integrated circuits.
- Setting up a semiconductor foundry requires a huge investment.
- Designing integrated circuits is done in semiconductor foundries.

No, the answer is incorrect.
Score: 0
Accepted Answers:
Designing integrated circuits is done in semiconductor foundries.

5) Which among the following design styles require maximum design effort: *1 point*

- Standard cell-based design
- Full-custom design
- Gate array-based-design
- FPGA-based design

No, the answer is incorrect.
Score: 0
Accepted Answers:
Full-custom design

6) Which among the following is a variable cost in standard cell-based design: *1 point*

- Design cost
- Cost of mask
- Cost of design tools
- Cost of die

No, the answer is incorrect.
Score: 0
Accepted Answers:
Cost of die

7) Which of the following representation of a design has the highest degree of abstraction: *1 point*

- As RTL in Verilog
- As an algorithm in the programming language C
- As connection of logic gates in Verilog
- As the layout of transistors and interconnects.

No, the answer is incorrect.
Score: 0
Accepted Answers:
As an algorithm in the programming language C

8) For which of the following figures of merit, dedicated hardware can be better than the same function implemented as software running on a general-purpose processor: *1 point*

- Risk due to bug
- Development time
- Speed
- Flexibility in implementation and customization

No, the answer is incorrect.
Score: 0
Accepted Answers:
Speed

9) What is the purpose of carrying out hardware-software partitioning for a system? *1 point*

- It gets rid of all the software in the system.
- It chooses an optimal combination of hardware and software to implement the system.
- It gets rid of all the hardware in the system.
- It compiles the software of the system.

No, the answer is incorrect.
Score: 0
Accepted Answers:
It chooses an optimal combination of hardware and software to implement the system.

10) Which of the following statements **best describes** the purpose of behavior synthesis in VLSI design flow? *1 point*

- It converts an algorithm to an equivalent design in RTL.
- It converts an algorithm to a binary executable.
- It converts an algorithm into sequences of zeros and ones.
- It converts an algorithm to an assembly language code.

No, the answer is incorrect.
Score: 0
Accepted Answers:
It converts an algorithm to an equivalent design in RTL.

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- Overview of VLSI Design Flow - III
- Overview of VLSI Design Flow - IV
- Overview of VLSI Design Flow - V
- Overview of VLSI Design Flow - VI
- Introduction to TCL
- Lecture Materials
- Quiz: Week 2: Assignment 2
- Week 2 Feedback Form : VLSI Design Flow: RTL to GDS
- Week 2: Solution

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Week 2: Assignment 2

The due date for submitting this assignment has passed.

Due on 2023-08-09, 23:59 IST.

As per our records you have not submitted this assignment.

1) Which of the following statements **best describes** the purpose of logic synthesis? *1 point*

- It converts an RTL to an equivalent netlist (interconnection of logic gates)
- It converts an RTL to an equivalent algorithm in C
- It converts an RTL to a technology library in Liberty
- It converts an RTL to a constraint in SDC

No, the answer is incorrect.
Score: 0
Accepted Answers:
It converts an RTL to an equivalent netlist (interconnection of logic gates)

2) Which of the following information is typically **NOT** needed by a logic synthesis tool: *1 point*

- RTL design [in Verilog or VHDL]
- Constraints [in Synopsys Design Constraint (SDC) format]
- Simulation output [in Value Change Dump (VCD) format]
- Technology Library [in Liberty format]

No, the answer is incorrect.
Score: 0
Accepted Answers:
Simulation output [in Value Change Dump (VCD) format]

3) What is the interface of a library cell or cell instance of a design called *1 point*

- Via
- Pin
- Jack
- Port

No, the answer is incorrect.
Score: 0
Accepted Answers:
Pin

4) Which of the following statements **best describes** the purpose of physical design in VLSI design flow? *1 point*

- It creates a library for a given netlist
- It creates an algorithm for a given netlist
- It creates a testbench for a given netlist
- It creates a layout for a given netlist

No, the answer is incorrect.
Score: 0
Accepted Answers:
It creates a layout for a given netlist

5) Which of the following tasks is **not a part** of the physical design: *1 point*

- Placement
- Routing
- Clock tree synthesis
- RTL synthesis

No, the answer is incorrect.
Score: 0
Accepted Answers:
RTL synthesis

6) Which of the following statements **best describes** the purpose of verification in VLSI design flow? *1 point*

- It ensures that the design makes a customer happy
- It ensures that the design works as per the given functionality
- It ensures that the design attracts many customers
- It ensures that the design makes a lot of profit

No, the answer is incorrect.
Score: 0
Accepted Answers:
It ensures that the design works as per the given functionality

7) Which of the following tasks is not a kind of design verification task: *1 point*

- Floorplanning
- Static Timing Analysis
- Combinational Equivalence Checking
- RTL Simulation

No, the answer is incorrect.
Score: 0
Accepted Answers:
Floorplanning

8) Which of the following statements **best describes** the purpose of the manufacturing test? *1 point*

- It ensures that the design is made as per the functionality
- It ensures that the yield of the process is high
- It ensures that no one can make a pirated chip out of the manufactured chip
- It checks whether the manufactured chip has a fatal defect.

No, the answer is incorrect.
Score: 0
Accepted Answers:
It checks whether the manufactured chip has a fatal defect.

9) Yield for a chip reduces when: *1 point*

- Defect density is decreased by better process control
- Probability of introducing random defects during fabrication is decreased
- The area of a die is increased
- The area of a die is decreased

No, the answer is incorrect.
Score: 0
Accepted Answers:
The area of a die is increased

10) Which of the following techniques is **NOT** a resolution enhancement technique: *1 point*

- Optical proximity correction
- Dummy metal fill
- Double-patterning
- Multi-patterning

No, the answer is incorrect.
Score: 0
Accepted Answers:
Dummy metal fill

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Course outline

How does an NPTEL online course work?

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- Hardware Modeling: Introduction to Verilog-I
- Hardware Modeling: Introducton to Verilog-II
- Functional Verification using Simulation
- High-level synthesis using Bambu - Tutorial 3
- Lecture Materials
- Quiz: Week 3: Assignment 3
- Week 3 Feedback Form : VLSI Design Flow: RTL to GDS
- Week 3: Solution

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Week 3: Assignment 3

The due date for submitting this assignment has passed.

Due on 2023-08-16, 23:59 IST.

As per our records you have not submitted this assignment.

1) How will the number -8'd4 (in Verilog) be internally represented as bits? *1 point*

- 1111 1111
- 1111 1110
- 1111 1100
- 1111 0000

No, the answer is incorrect.
Score: 0
Accepted Answers:
1111 1100

2) Which of the following data types can store value during Verilog simulation? *1 point*

- wire
- wand
- wor
- reg

No, the answer is incorrect.
Score: 0
Accepted Answers:
reg

3) The variable A contains the bit sequence (1011). The variable Y is obtained by applying the reduction operator: Y=|A
What will be the final value of Y? *1 point*

- 0
- 1
- X
- Z

No, the answer is incorrect.
Score: 0
Accepted Answers:
1

4) Which of the following statements about non-blocking assignments in Verilog is **incorrect**? *1 point*

- The execution of the next statement is done after the current non-blocking assignment is fully executed
- The symbol for non-blocking assignment is <=
- The execution of the next statement is not blocked until the current non-blocking assignment is executed
- The RHS of a non-blocking assignment can be a Verilog expression

No, the answer is incorrect.
Score: 0
Accepted Answers:
The execution of the next statement is done after the current non-blocking assignment is fully executed

5) Which of the following statements is **incorrect** about a Verilog function? *1 point*

- A function can have one or more inputs
- A function will have one output
- A function can call another function
- A function can call another task

No, the answer is incorrect.
Score: 0
Accepted Answers:
A function can call another task

6) Which of the following tasks is typically **NOT** performed by a testbench in simulation-based RTL verification: *1 point*

- Interact with the simulator
- Applies stimulus
- Observes the response
- Perform logic synthesis

No, the answer is incorrect.
Score: 0
Accepted Answers:
Perform logic synthesis

7) Which of the following statements is **incorrect** about code coverage: *1 point*

- Help identify sections of DUV's source code that did not execute during verification.
- Help perform timing analysis of the design
- Helps monitor the progress of the verification effort
- Helps assess the quality of verification

No, the answer is incorrect.
Score: 0
Accepted Answers:
Help perform timing analysis of the design

8) An event-based RTL simulator keeps the events ordered in the event queue according to: *1 point*

- Speed of the processor on which simulation is run
- Randomly
- Simulation time of the event
- Delay of circuit elements in the synthesized netlist

No, the answer is incorrect.
Score: 0
Accepted Answers:
Simulation time of the event

9) Which of the following Verilog constructs is processed last in the stratified Verilog event queue *1 point*

- continuous assign statement
- blocking assignment
- LHS of non-blocking assignment
- RHS of non-blocking assignment

No, the answer is incorrect.
Score: 0
Accepted Answers:
LHS of non-blocking assignment

10) Which of the following statements is **incorrect** about race in RTL simulation *1 point*

- Race in RTL simulation signifies that the circuit is operating very fast
- Due to race, we can obtain different results when the order of execution of two or more statements is changed (as permitted by IEEE standard)
- Races in RTL simulation can be difficult to debug
- Races can be avoided by following some guidelines in RTL coding

No, the answer is incorrect.
Score: 0
Accepted Answers:
Race in RTL simulation signifies that the circuit is operating very fast

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Course outline

How does an NPTEL online course work?

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Week 4

- RTL Synthesis- Part I
- RTL Synthesis- Part II
- Logic Optimization: Part I
- Simulation-based Verification using Icarus
- Lecture Materials
- Quiz: Week 4: Assignment 4
- Week 4 Feedback Form : VLSI Design Flow: RTL to GDS
- Week 4: Solution

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Week 4: Assignment 4

The due date for submitting this assignment has passed.

Due on 2023-08-23, 23:59 IST.

As per our records you have not submitted this assignment.

1) Which of the following tasks is typically **not a part** of RTL synthesis? *1 point*

- Parsing
- Model checking
- Elaboration
- Translation to logic gates

No, the answer is incorrect.
Score: 0
Accepted Answers:
Model checking

2) How will a parameterized module (with a parameter named WIDTH) and having a port named A (with size WIDTH) be processed during elaboration in RTL synthesis? *1 point*

- Only one module with a fixed size for port A will be created for all WIDTH values.
- Port A will be eliminated from the parameterized module.
- Separate modules with different size of port A for each distinct value of WIDTH will be created.
- The parameterized module will be ignored during elaboration.

No, the answer is incorrect.
Score: 0
Accepted Answers:
Separate modules with different size of port A for each distinct value of WIDTH will be created.

3) Which of the following Verilog construct is typically **non-synthesizable**? *1 point*

- assign statement
- always block
- case statement
- $display statement

No, the answer is incorrect.
Score: 0
Accepted Answers:
$display statement

4) Level-sensitive always block will be synthesized to: *1 point*

- Flip-flop or latches
- Flip-flop or combinational circuit elements
- Combinational circuit elements or latches
- Only latches

No, the answer is incorrect.
Score: 0
Accepted Answers:
Combinational circuit elements or latches

5) What will be the synthesis result of the following Verilog code? *1 point*

```
module top2(in1, clk, out1);
    input in1, clk;
    output out1;
    reg reg1, reg2, reg3, out1;

    always @ ( posedge clk)
    begin
        reg1 <= in1;
        reg2 <= reg1;
        reg3 <= reg2;
        out1 <= reg3;
    end
endmodule
```

- 4-bit Ripple counter
- 4-bit Shift register
- One flip-flop
- 4-bit ripple carry adder

No, the answer is incorrect.
Score: 0
Accepted Answers:
4-bit Shift register

6) What will be the synthesis result of the following piece of Verilog code: *1 point*

```
always @(*) begin
    if (en==1'b1) out2 = c;
    else out2 = out2;
end
```

- One AND gate
- One D-Latch
- One D flip-flop
- One Full-Adder

No, the answer is incorrect.
Score: 0
Accepted Answers:
One D-Latch

7) What is the primary motivation for resource sharing in RTL synthesis: *1 point*

- Save area
- Increase the maximum operable frequency of the circuit.
- Increase the testability of the circuit.
- Improve the routability of the circuit.

No, the answer is incorrect.
Score: 0
Accepted Answers:
Save area

8) An implicant of an incompletely specified Boolean function is a cube whose all corners are on: *1 point*

- either ON-set or OFF-set
- either ON-set or DC-set
- either OFF-set or DC-set
- Only ON-set

No, the answer is incorrect.
Score: 0
Accepted Answers:
either ON-set or DC-set

9) Which of the following statements is **incorrect** regarding two-level logic minimization *1 point*

- The prime cover consists of only prime implicants.
- There exists a minimum cover consisting of only prime implicants.
- There exists a minimum cover consisting of only essential prime implicants.
- All the essential prime implicants must be included in a prime cover.

No, the answer is incorrect.
Score: 0
Accepted Answers:
There exists a minimum cover consisting of only essential prime implicants.

10) Which of the following statements is incorrect about the two-level **heuristic** minimizer *1 point*

- The cover size obtained using a heuristic minimizer can exceed the minimum cover.
- Two-level heuristic minimizer typically produces solutions quicker than two-level exact minimization for large problems.
- A heuristic minimizer typically tries to obtain a minimal cover by applying operators iteratively until the solution improves.
- The cover produced by a heuristic minimizer is guaranteed to be the minimum cover.

No, the answer is incorrect.
Score: 0
Accepted Answers:
The cover produced by a heuristic minimizer is guaranteed to be the minimum cover.

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Course outline

How does an NPTEL online course work?

Week 0

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- Logic Optimization: Part II
- Logic Optimization: Part III
- Formal Verification-I
- Logic Synthesis using Yosys
- Lecture Materials 5
- Quiz: Week 5: Assignment 5
- Week 5 Feedback Form : VLSI Design Flow: RTL to GDS
- Week 5: Solution

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Week 5: Assignment 5

The due date for submitting this assignment has passed.

Due on 2023-08-30, 23:59 IST.

As per our records you have not submitted this assignment.

1) Which of the following statements is **correct** for multilevel logic? *1 point*

- A multilevel logic representation will always produce a minimum delay between the input and the output.
- As the number of levels in a multilevel logic increases, the delay between the input and the output decreases.
- A multilevel logic can be transformed to explore various delay-area trade-offs.
- A multilevel logic representation cannot be transformed into a two-level logic representation.

No, the answer is incorrect.
Score: 0
Accepted Answers:
A multilevel logic can be transformed to explore various delay-area trade-offs.

2) Which of the following statements about the algebraic model in multilevel logic optimization is **incorrect**? *1 point*

- An algebraic model neglects some Boolean properties of the local functions.
- An algebraic model treats a variable and its complement as separate variables.
- Efficient algorithms can be designed to carry out division in an algebraic model.
- An algebraic model can reduce the Boolean expression $(a + a')$ to 1

No, the answer is incorrect.
Score: 0
Accepted Answers:
An algebraic model can reduce the Boolean expression $(a + a')$ to 1

3) A minterm is found to be Don't Care (DC) during logic optimization. Which of the following statements **best describes** the behavior of the logic synthesis tool? *1 point*

- A logic synthesis tool will implement a circuit that can produce either 0 or 1 for this minterm, depending on which of these values reduces the cost.
- A logic synthesis tool will implement a circuit that always produces 0 for this minterm.
- A logic synthesis tool will implement a circuit that always produces 1 for this minterm.
- A logic synthesis tool will implement a circuit that always produces high impedance (Z value) for this minterm.

No, the answer is incorrect.
Score: 0
Accepted Answers:
A logic synthesis tool will implement a circuit that can produce either 0 or 1 for this minterm, depending on which of these values reduces the cost.

4) Which of the following statements is **incorrect** for a typical sequential circuit implementation of a finite state machine (FSM): *1 point*

- The combination of the Q-pin value of flip-flops represents states.
- The transition function is implemented using combinational logic.
- Reducing the number of states in an FSM can never reduce the size of the circuit implementation.
- State minimization and optimal state encoding can reduce the size of the circuit implementation.

No, the answer is incorrect.
Score: 0
Accepted Answers:
Reducing the number of states in an FSM can never reduce the size of the circuit implementation.

5) Which of the following statements is **incorrect** about state minimization in a finite state machine (FSM): *1 point*

- State minimization attempts to derive an FSM with the minimum number of states and exhibiting the same behavior as the original FSM.
- State minimization can reduce the number of flip-flops in FSM implementation.
- State minimization relies on retaining any one of the equivalent states and removing others.
- State minimization will change the functionality/behavior of an FSM and should not be done.

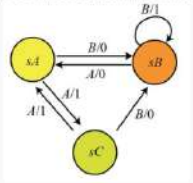
No, the answer is incorrect.
Score: 0
Accepted Answers:
State minimization will change the functionality/behavior of an FSM and should not be done.

6) Which of the following state encoding schemes for a finite state machine (FSM) reserves one bit for each state? *1 point*

- One-hot Encoding
- Binary Encoding
- Gray Encoding
- Zero-shot Encoding

No, the answer is incorrect.
Score: 0
Accepted Answers:
One-hot Encoding

7) Which of the following pairs of states are equivalent in the finite state machine (FSM) whose state diagram is shown below? *1 point*



- sA and sB
- sA and sC
- sB and sC
- None of the states are equivalent

No, the answer is incorrect.
Score: 0
Accepted Answers:
sA and sC

8) Formal verification techniques rely on the following: *1 point*

- Simulation
- Mathematical Proof
- Random Guesses
- Statistical Inference

No, the answer is incorrect.
Score: 0
Accepted Answers:
Mathematical Proof

9) A Reduced Ordered Binary Decision Diagram (ROBDD) of a Boolean function: *1 point*

- Is a canonical representation
- Is the same as a truth table
- Always represents a Boolean function compactly.
- Has size independent of the variable order

No, the answer is incorrect.
Score: 0
Accepted Answers:
Is a canonical representation

10) What can we say about the Reduced Ordered Binary Decision Diagram (ROBDD) representation of a Boolean function $f = ab(a' + b')$ [where a and b are Boolean variables]? *1 point*

- It will contain one non-terminal vertex and two leaf terminals.
- It will contain two non-terminal vertex and two leaf terminals.
- It will contain only one leaf terminal with the label 1
- It will contain only one leaf terminal with the label 0

No, the answer is incorrect.
Score: 0
Accepted Answers:
It will contain only one leaf terminal with the label 0

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Course outline

How does an NPTEL online course work?

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- Formal Verification-II
- Formal Verification-III
- Formal Verification-IV
- Technology Library
- Logic Optimization using Yosys
- Lecture Materials 6
- Quiz: Week 6: Assignment 6
- Week 6 Feedback Form : VLSI Design Flow: RTL to GDS
- Week 6: Solution

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Week 6: Assignment 6

The due date for submitting this assignment has passed.

Due on 2023-09-06, 23:59 IST.

As per our records you have not submitted this assignment.

1) Assume that a given Boolean function is $y = f(x_1, x_2, x_3, \ldots, x_n)$ where $x_1, x_2, x_3, \ldots, x_n$ are Boolean variables. Which of the following statements describes the satisfiability problem **correctly**? **1 point**

- It determines if y can be evaluated to 0 by any assignment of variables $\{x_1, x_2, x_3, \ldots, x_n\}$.
- It determines if y can be evaluated to 0 by some assignment of variables $\{x_1, x_2, x_3, \ldots, x_n\}$.
- It determines if y can be evaluated to 1 by any assignment of variables $\{x_1, x_2, x_3, \ldots, x_n\}$.
- It determines if y can be evaluated to 1 by all assignments of variables $\{x_1, x_2, x_3, \ldots, x_n\}$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
It determines if y can be evaluated to 1 by any assignment of variables $\{x_1, x_2, x_3, \ldots, x_n\}$.

2) Assume that a given Boolean function is $y = f(x_1, x_2, x_3, \ldots, x_n)$ where $\{x_1, x_2, x_3, \ldots, x_n\}$ are Boolean variables. In **the worst case**, how many input combinations need to be tried by a SAT solver? **1 point**

- Constant number of combinations
- Linear in the number of variables *n*
- Polynomial in the number of variables *n*
- Exponential in the number of variables *n*

No, the answer is incorrect.
Score: 0
Accepted Answers:
Exponential in the number of variables n

3) Given a set of states S for a finite state machine (FSM), the problem of obtaining a set of states S' that can be reached in one step from S is known as: **1 point**

- Image computation
- Pre-image computation
- State encoding
- State minimization

No, the answer is incorrect.
Score: 0
Accepted Answers:
Image computation

4) Which of the following statements is **incorrect** for Bounded Model Checking (BMC): **1 point**

- BMC tries to obtain a counterexample of a finite length *n* (*n* is the number of clock cycles from the initial state).
- BMC lacks completeness because it considers counterexamples of a finite length.
- BMC is useful in finding bugs if a bug can be obtained in a few clock cycles from the initial state.
- BMC uses exhaustive simulation to find counterexamples of infinite length *n* (*n* is the number of clock cycles from the initial state)

No, the answer is incorrect.
Score: 0
Accepted Answers:
BMC uses exhaustive simulation to find counterexamples of infinite length n (n is the number of clock cycles from the initial state)

5) Which of the following tasks is **not a part** of Combinational Equivalence Checking (CEC): **1 point**

- Port and register matching
- Dynamic power analysis
- Miter creation
- Establishing equivalence of miters

No, the answer is incorrect.
Score: 0
Accepted Answers:
Dynamic power analysis

6) In which of the following cases port matching **can fail** during Combinational Equivalence Checking (CEC): **1 point**

- When some buffers are added to the netlist
- When a NAND gate is optimized using Boolean algebra
- When the size of an inverter is increased to fix load violation
- When an extra port is added to a design

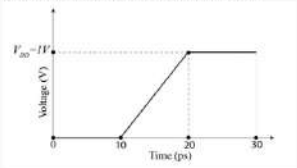
No, the answer is incorrect.
Score: 0
Accepted Answers:
When an extra port is added to a design

7) Technology libraries are typically stored in which of the following formats: **1 point**

- Standard Parasitic Exchange Format (SPEF)
- Synopsys Liberty Format
- Synopsys Design Constraint (SDC) Format
- Value Change Dump (VCD) Format

No, the answer is incorrect.
Score: 0
Accepted Answers:
Synopsys Liberty Format

8) The slew of the signal shown below computed using a 20-80 threshold is: **1 point**



- 2 ps
- 4 ps
- 6 ps
- 8 ps

No, the answer is incorrect.
Score: 0
Accepted Answers:
6 ps

9) The Non-linear Delay Model (NLDM) models the delay of a combinational circuit element as a function of: **1 point**

- Voltage and Temperature
- Process Parameter and Input Slew
- Input Slew and Output Load
- Voltage and driving resistance

No, the answer is incorrect.
Score: 0
Accepted Answers:
Input Slew and Output Load

10) The setup time of a flip-flop in a technology library is modelled as a function of which two variables? **1 point**

- The voltage at the D-pin and Load at the Q-pin
- Input Slew at the D-pin and Input Slew at the Clock Pin
- Temperature and Input Slew at the D-pin
- Input Slew at the D-pin and Load at the Q-pin

No, the answer is incorrect.
Score: 0
Accepted Answers:
Input Slew at the D-pin and Input Slew at the Clock Pin

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How does an NPTEL online course work?

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- Static Timing Analysis- I
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Week 7: Assignment 7

The due date for submitting this assignment has passed.

Due on 2023-09-13, 23:59 IST.

As per our records you have not submitted this assignment.

1) In a typical synchronous design, when will the data launched by a flip-flop at a given clock edge be captured by the adjacent flip-flop in its fanout? *1 point*

- At the same clock edge
- At the next clock edge
- At the next-to-next clock edge
- Will never be captured

No, the answer is incorrect.
Score: 0
Accepted Answers:
At the next clock edge

2) The arrival time of the data at the capture flip-flop does **NOT** depend on the following: *1 point*

- Setup time of the capture flip-flop
- Clock-to-Q delay of the launch flip-flop
- Data path delay
- Delay on the clock path of the launch flip-flop

No, the answer is incorrect.
Score: 0
Accepted Answers:
Setup time of the capture flip-flop

3) The hold requirement constraint does **NOT** depend on which of the following quantities in a typical synchronous circuit? *1 point*

- Clock Period
- Data Path Delay
- Delay on the clock path of the launch flip-flop
- Delay on the clock path of the capture flip-flop

No, the answer is incorrect.
Score: 0
Accepted Answers:
Clock Period

4) The delay calculation for a given stage performed during Static Timing Analysis will **NOT** depend on the following: *1 point*

- Interconnect model
- Driver model of the driving cell
- Yield model for the technology
- Receiver model of the driven cell

No, the answer is incorrect.
Score: 0
Accepted Answers:
Yield model for the technology

5) Which of the following is **NOT** considered in creating scenarios for multi-mode multi-corner (MMMC) analysis? *1 point*

- PVT corners
- RC corners
- Floorplan Corners
- Design Modes

No, the answer is incorrect.
Score: 0
Accepted Answers:
Floorplan Corners

6) We have defined a ***late derating factor*** of 1.1.
For the early analysis, which of the following quantities should be multiplied by the above derating factor to make the analysis more pessimistic? *1 point*

- Data path delay
- Clock-to-Q delay of the launch flip-flop
- Delay of the clock launch path
- Delay of the clock capture path

No, the answer is incorrect.
Score: 0
Accepted Answers:
Delay of the clock capture path

7) Consider the AND gate shown below: *1 point*



The delay and the output slew of different arcs are shown below:

Arc	Delay (ps)	Output Slew at pin Z (ps)
A→Z	10	20
B→Z	20	5
C→Z	30	10

The arrival time at different input pins are as follows:

Pin	Arrival Time (ps)
A	5
B	6
C	12

What will be the arrival time and the output slew at the pin Z when **Graph Based Analysis (GBA)** is performed for the setup check or late-mode check?

- Arrival Time = 42 ps Output Slew = 10 ps
- Arrival Time = 15 ps Output Slew = 20 ps
- Arrival Time = 42 ps Output Slew = 20 ps
- Arrival Time = 15 ps Output Slew = 10 ps

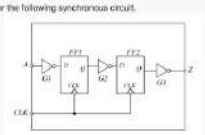
No, the answer is incorrect.
Score: 0
Accepted Answers:
Arrival Time = 42 ps Output Slew = 20 ps

8) For the same question above, what will be the arrival time and the output slew at the pin Z when **Path Based Analysis (PBA)** is performed for the setup check or late-mode check through the arc $B \rightarrow Z$? *1 point*

- Arrival Time = 26 ps Output Slew = 20 ps
- Arrival Time = 26 ps Output Slew = 5 ps
- Arrival Time = 42 ps Output Slew = 20 ps
- Arrival Time = 42 ps Output Slew = 5 ps

No, the answer is incorrect.
Score: 0
Accepted Answers:
Arrival Time = 26 ps Output Slew = 5 ps

9) Consider the following synchronous circuit. *1 point*



The following attributes are valid for the flip-flops FF1 and FF2: setup time=45 ps, hold time=10 ps, and CLK-to-Q delay=20 ps. The delay of each inverter is 60 ps. Ignore the wire delay.
Assume that the period of the clock is 1000 ps. What is the **setup slack** at the timing end-point FF2/D?

- 885 ps
- 900 ps
- 0 ps
- -10 ps

No, the answer is incorrect.
Score: 0
Accepted Answers:
885 ps

10) For the same question above, what is the **hold slack** at the timing end-point FF2/D? *1 point*

- 70 ps
- 60 ps
- -60 ps
- -70 ps

No, the answer is incorrect.
Score: 0
Accepted Answers:
60 ps

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- Technology Library and Constraints
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Week 8: Assignment 8

The due date for submitting this assignment has passed.

Due on 2023-09-20, 23:59 IST.

As per our records you have not submitted this assignment.

1) The SDC command to define a primary clock is *1 point*

- create_toggle
- create_period
- create_clock
- start_clock

No, the answer is incorrect.
Score: 0
Accepted Answers:
create_clock

2) The SDC command to define delay at the input port is *1 point*

- set_delay_my_port
- set_input_delay
- set_output_delay
- set_delay_my_input

No, the answer is incorrect.
Score: 0
Accepted Answers:
set_input_delay

3) The SDC command to add extra margin to the timing requirements is *1 point*

- set_clock_uncertainty
- set_my_margin
- set_extra_margin
- set_design_extra_margin

No, the answer is incorrect.
Score: 0
Accepted Answers:
set_clock_uncertainty

4) The setup check is done at the output port when which of the following constraints is applied at that port? *1 point*

- set_input_delay
- set_output_delay
- set_false_path
- set_input_transition

No, the answer is incorrect.
Score: 0
Accepted Answers:
set_output_delay

5) Which of the following statements **BEST** describes the purpose of technology mapping in logic synthesis: *1 point*

- It converts a given generic gate netlist to a netlist consisting of standard cells such that functionality is preserved, and the given cost is minimized.
- It converts a given generic gate netlist to a netlist consisting of standard cells such that an arbitrary functionality is realized, and the given cost is minimized.
- It converts a given generic gate netlist to a netlist consisting of standard cells such that functionality is preserved, and the given cost is maximized.
- It converts a given generic gate netlist to a netlist consisting of transistors such that an arbitrary functionality is realized, and the given cost is minimized.

No, the answer is incorrect.
Score: 0
Accepted Answers:
It converts a given generic gate netlist to a netlist consisting of standard cells such that functionality is preserved, and the given cost is minimized.

6) Which of the following optimization techniques is typically **NOT** targeted for improving timing: *1 point*

- Resizing
- Power Gating
- Restructuring
- Retiming

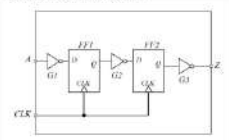
No, the answer is incorrect.
Score: 0
Accepted Answers:
Power Gating

7) Which of the following optimization techniques involves changing logic circuit elements across a flip-flop: *1 point*

- Resizing
- Retiming
- Restructuring
- Fanout optimization

No, the answer is incorrect.
Score: 0
Accepted Answers:
Retiming

8) Consider the following synchronous circuit. *1 point*



The following attributes are valid for the flip-flops FF1 and FF2: setup time=45 ps, hold time=10 ps, and CLK-to-Q delay=20 ps. The delay of each inverter is 50 ps. Ignore the wire delay.
Assume that we have define the following constraints in the SDC file (all time units are in picoseconds):

```
create_clock -name CLK -period 1000 [get_ports CLK]
set_input_delay -clock [get_clocks CLK] 150 [get_ports A]
set_output_delay -clock [get_clocks CLK] 125 [get_ports Z]
```

What is the **setup slack** at the timing end-point FF1/D?

- 755 ps
- 700 ps
- 850 ps
- 905 ps

No, the answer is incorrect.
Score: 0
Accepted Answers:
755 ps

9) For the question above, what is the **hold slack** at the timing end-point FF1/D? *1 point*

- 200 ps
- 190 ps
- 0 ps
- -190 ps

No, the answer is incorrect.
Score: 0
Accepted Answers:
190 ps

10) For the question above, what is the **setup slack** at the timing end-point port Z? *1 point*

- 800 ps
- 900 ps
- 805 ps
- 905 ps

No, the answer is incorrect.
Score: 0
Accepted Answers:
805 ps

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- Basic Concepts of DFT
- Scan Design Flow
- Power Analysis using OpenSTA
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Week 9: Assignment 9

The due date for submitting this assignment has passed.

Due on 2023-09-27, 23:59 IST.

As per our records you have not submitted this assignment.

1) The power dissipated in a CMOS circuit when it is not performing active computation is known as: *1 point*

- Switching power dissipation
- Dynamic power dissipation
- Static power dissipation
- Short circuit power dissipation

No, the answer is incorrect.
Score: 0
Accepted Answers:
Static power dissipation

2) The switching power dissipation of a CMOS combinational circuit element increases under which of the following conditions: *1 point*

- The period of the clock is decreased.
- The supply voltage is decreased.
- The activity at the input of the circuit element is decreased.
- The capacitive load driven by that circuit element is decreased.

No, the answer is incorrect.
Score: 0
Accepted Answers:
The period of the clock is decreased.

3) The internal power in a technology library's non-linear power model (NLPM) is typically defined as a function of which of the following quantities? *1 point*

- Switching activity and supply voltage
- Input slew and output load
- Clock frequency and supply voltage
- Switching activity and output load

No, the answer is incorrect.
Score: 0
Accepted Answers:
Input slew and output load

4) Which of the following is **NOT** a technique to estimate a signal's activity in power analysis? *1 point*

- Vector-based Technique
- Probabilistic Technique
- Resizing Technique
- Vector-less Technique

No, the answer is incorrect.
Score: 0
Accepted Answers:
Resizing Technique

5) Which of the following techniques is typically **NOT** a technique to reduce power dissipation in a circuit? *1 point*

- Clock Gating
- Redundant Via Insertion
- Power Gating
- Dynamic Voltage Frequency Scaling (DVFS)

No, the answer is incorrect.
Score: 0
Accepted Answers:
Redundant Via Insertion

6) Which of the following statements best describes the problem of using a simple AND gate for clock gating? *1 point*

- In some cases, it can cause the output of the AND gate to get stuck to one, and the circuit will go into a metastable state.
- It can create glitches at the output of the AND gate, and this glitch can trigger a flip-flop in its fanout.
- It cannot switch off the power supply and not be able to save any power.
- It can cause a significant voltage drop in the supply lines, increasing the delay of other circuit elements.

No, the answer is incorrect.
Score: 0
Accepted Answers:
It can create glitches at the output of the AND gate, and this glitch can trigger a flip-flop in its fanout.

7) Which of the following is NOT the purpose of employing a fault model in design for test (DFT) tasks? *1 point*

- It allows us to analyze the impact of a defect using logic or circuit analysis techniques.
- It helps us reduce the number of defects during integrated circuit fabrication.
- It allows us to derive test patterns algorithmically for detecting a given fault.
- It helps us transform the problem of defect detection into the problem of fault detection.

No, the answer is incorrect.
Score: 0
Accepted Answers:
It helps us reduce the number of defects during integrated circuit fabrication.

8) From the perspective of design for test (DFT), which of the following statements **best describes** a test vector: *1 point*

- Any input pattern, or sequence of input patterns, that produces a 1 for a faulty circuit and a fault-free circuit
- Any input pattern, or sequence of input patterns, that produces a 0 for a faulty circuit and a fault-free circuit
- Any input pattern, or sequence of input patterns, that produces a matching output response for a faulty circuit and a fault-free circuit
- Any input pattern, or sequence of input patterns, that produces a different output response for a faulty circuit and a fault-free circuit

No, the answer is incorrect.
Score: 0
Accepted Answers:
Any input pattern, or sequence of input patterns, that produces a different output response for a faulty circuit and a fault-free circuit

9) In scan-based testing methodology, we replace a D flip-flop with a scan cell. What is the motivation for this: *1 point*

- It decreases the delay and area of the circuit.
- It increases the manufacturability of the circuit.
- It increases the controllability and observability of a circuit.
- It increases the performance and reliability of a circuit.

No, the answer is incorrect.
Score: 0
Accepted Answers:
It increases the controllability and observability of a circuit.

10) In scan-based testing methodology, what is the correct value at the SE (scan enable) port for the shift and capture mode, respectively? *1 point*

- Shift SE=0, Capture SE=0
- Shift SE=0, Capture SE=1
- Shift SE=1, Capture SE=0
- Shift SE=1, Capture SE=1

No, the answer is incorrect.
Score: 0
Accepted Answers:
Shift SE=1, Capture SE=0

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Course outline

How does an NPTEL online course work?

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- Automatic Test Pattern Generation (ATPG)
- Built-in Self Test (BIST)
- Basic Concepts for Physical Design - I
- Basic Concepts for Physical Design - II
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Week 10: Assignment 10

The due date for submitting this assignment has passed.

Due on 2023-10-04, 23:59 IST.

As per our records you have not submitted this assignment.

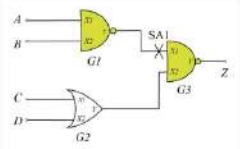
1) The non-controlling value of a two-input XOR gate is: *1 point*

- 0
- 1
- Both 0 and 1
- Neither 0 nor 1

No, the answer is incorrect.
Score: 0

Accepted Answers:
Both 0 and 1

2) For the circuit shown below, the test pattern to detect SA1 at G3/X1 is: *1 point*



- A=1 B=1 C=0 D=1
- A=0 B=0 C=0 D=1
- A=1 B=1 C=0 D=0
- A=0 B=1 C=0 D=1

No, the answer is incorrect.
Score: 0

Accepted Answers:
A=1 B=1 C=0 D=1

3) Which of the following components is typically **NOT** a part of Built-in Self-test (BIST) infrastructure? *1 point*

- Random Test Pattern Generator
- Test Response Analyzer
- JPEG Encoder
- BIST controller

No, the answer is incorrect.
Score: 0

Accepted Answers:
JPEG Encoder

4) The patterns obtained using Linear Feedback Shift Register (LFSR) are: *1 point*

- Finite, repeatable, and pseudo-random
- Infinite, repeatable, and truly random
- Finite, non-repeatable, and truly random
- Infinite, non-repeatable, and pseudo-random

No, the answer is incorrect.
Score: 0

Accepted Answers:
Finite, repeatable, and pseudo-random

5) What is aliasing in signature analysis from the perspective of Built-in Self-test (BIST) *1 point*

- Matching of signature of "good" circuit and "faulty" circuit
- Matching of signature of the designer and the signature of the engineer who is fabricating
- Overlapping of frequency components in automatic test equipment
- Distortion in the outline of the image of the chip taken during testing

No, the answer is incorrect.
Score: 0

Accepted Answers:
Matching of signature of "good" circuit and "faulty" circuit

6) What is the primary purpose of ion implantation in integrated circuit fabrication? *1 point*

- Etching the photoresist
- Doping the semiconductor
- Applying a coating of metal
- Growing a thick layer of oxide

No, the answer is incorrect.
Score: 0

Accepted Answers:
Doping the semiconductor

7) A metal interconnect line in an integrated circuit has a sheet resistance of 0.06 Ohms per square. The length of the interconnect is 10 microns and the width is 0.01 microns. What is the resistance of the interconnect? *1 point*

- 0.6 Ohms
- 6 Ohms
- 60 Ohms
- 600 Ohms

No, the answer is incorrect.
Score: 0

Accepted Answers:
60 Ohms

8) The lateral capacitance between two adjacent interconnects running parallelly in the same layer will increase under which of the following conditions? *1 point*

- The dielectric constant of the insulator between them is decreased.
- The sidewall area of the interconnects is increased.
- The distance between the interconnects is increased.
- The overlapping length between them is decreased.

No, the answer is incorrect.
Score: 0

Accepted Answers:
The sidewall area of the interconnects is increased.

9) Which of the following phenomenon is typically NOT a manifestation of signal integrity issues in an integrated circuit? *1 point*

- Increase in delay from the base delay.
- Decrease in delay from the base delay.
- Introduction of a defect in a manufactured chip.
- Glitches in a signal within an integrated circuit.

No, the answer is incorrect.
Score: 0

Accepted Answers:
Introduction of a defect in a manufactured chip.

10) Which of the following files contain the information about the available layers and vias, their properties for a given technology *1 point*

- Verilog design files
- Testbench
- SDC files
- LEF files

No, the answer is incorrect.
Score: 0

Accepted Answers:
LEF files

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Week 11: Assignment 11

The due date for submitting this assignment has passed.

Due on 2023-10-11, 23:59 IST.

As per our records you have not submitted this assignment.

1) Which of the following tasks is **NOT** a part of hierarchical design implementation methodologies: *1 point*

- Partitioning
- Budgeting
- Top-level Assembly
- Post-silicon validation

No, the answer is incorrect.
Score: 0
Accepted Answers:
Post-silicon validation

2) Which of the following statements does **NOT** describe the **advantages** of hierarchical design implementation methodologies compared to flat design implementation methodologies for physical design: *1 point*

- The physical implementation tools need to handle smaller problems for various blocks.
- Different teams can concurrently implement multiple blocks.
- Overall design time can reduce due to the parallel implementation of blocks.
- The final figures of merit can suffer in hierarchical design implementation.

No, the answer is incorrect.
Score: 0
Accepted Answers:
The final figures of merit can suffer in hierarchical design implementation.

3) Which of the following functions is typically **NOT** provided by an IO cell in an integrated circuit: *1 point*

- Arithmetic operation
- Increase drive capability
- Voltage transformation
- Protection against Electrostatic Discharge (ESD)

No, the answer is incorrect.
Score: 0
Accepted Answers:
Arithmetic operation

4) The advantage of creating standard cell rows oriented in alternating 180-degree rotation is: *1 point*

- It allows us to obtain the correct logical functionality.
- It allows us to share the power and ground lines for adjacent rows.
- It removes glitches in the circuit induced by signal integrity issues.
- It eases the task of clock gating and power gating to save switching power dissipation.

No, the answer is incorrect.
Score: 0
Accepted Answers:
It allows us to share the power and ground lines for adjacent rows.

5) IR drop in the power lines can be reduced typically by *1 point*

- Increasing the resistance of the power supply lines.
- Using the higher levels of metal layers.
- Decreasing the width of the power supply lines.
- Using lower levels of metal layers.

No, the answer is incorrect.
Score: 0
Accepted Answers:
Using the higher levels of metal layers.

6) The phenomenon of transport of metal mass in the direction opposite to current flow due to the flow of unidirectional current is known as: *1 point*

- Implantation
- Diffusion
- Electromigration
- Ionization

No, the answer is incorrect.
Score: 0
Accepted Answers:
Electromigration

7) A capacitor or cell inserted between the power and ground lines to mitigate the problem of IR drop hotspots is known as *1 point*

- Dying Capacitor (Dead cell)
- Demanding Capacitor (Dem cell)
- Supplying Capacitor (Super cell)
- Decoupling Capacitor (Decap Cell)

No, the answer is incorrect.
Score: 0
Accepted Answers:
Decoupling Capacitor (Decap Cell)

8) Why do we need to carry out legalization after global placement during standard cell placement? *1 point*

- To avoid legal proceedings against designer
- To remove overlap among standard cells
- To make incremental changes in the functionality
- To reduce static power dissipation

No, the answer is incorrect.
Score: 0
Accepted Answers:
To remove overlap among standard cells

9) The reduction in the wirelength of the scan chains after placement is typically accomplished by which of the following tasks: *1 point*

- Scan Chain Minimization
- Scan Chain Reordering
- Scan Chain Resizing
- Scan Chain Retiming

No, the answer is incorrect.
Score: 0
Accepted Answers:
Scan Chain Reordering

10) The extra cells that we put in an integrated circuit in anticipation of later use are known as: *1 point*

- Spare Cell
- Base cell
- Pad Cell
- IO Cell

No, the answer is incorrect.
Score: 0
Accepted Answers:
Spare Cell

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- Clock Tree Synthesis (CTS) and Routing
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- Week 12 Feedback Form : VLSI Design Flow: RTL to GDS
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Week 12: Assignment 12

The due date for submitting this assignment has passed.

Due on 2023-10-18, 23:59 IST.

As per our records you have not submitted this assignment.

1) Which of the following is a type of symmetric clock tree architecture? *1 point*

- B-tree
- H-tree
- S-tree
- P-tree

No, the answer is incorrect.
Score: 0
Accepted Answers:
H-tree

2) Which of the following statements does **NOT** explain the advantages of employing a mesh architecture over tree architecture for a global clock distribution network? *1 point*

- It allows us to obtain a very small skew in the clock distribution network.
- It has a decreased impact of process-induced variations.
- It is more robust to failure due to open circuit defects.
- It has less short-circuit power dissipation in the drivers of the clock network.

No, the answer is incorrect.
Score: 0
Accepted Answers:
It has less short-circuit power dissipation in the drivers of the clock network.

3) Clock skews are considered useful skews under which of the following conditions? *1 point*

- Skew can improve the testability of the circuit.
- Skew can increase the system's maximum operable frequency.
- Skew can reduce the defect in the manufactured circuit.
- Skew can enhance the manufacturability of the circuit.

No, the answer is incorrect.
Score: 0
Accepted Answers:
Skew can increase the system's maximum operable frequency.

4) Global Routing is the planning phase for which of the following tasks? *1 point*

- ECO
- Signoff
- Tape out
- Detailed Routing

No, the answer is incorrect.
Score: 0
Accepted Answers:
Detailed Routing

5) Global routing attempts to route each net with: *1 point*

- Maximum overflow
- Overflow of Zero
- Maximum congestion
- Maximum wirelength

No, the answer is incorrect.
Score: 0
Accepted Answers:
Overflow of Zero

6) Which of the following techniques is employed to tackle via defects *1 point*

- Addition of redundant vias
- Addition of integrated clock gating (ICG) cells
- Addition of spare cells
- Addition of redundant IO cells

No, the answer is incorrect.
Score: 0
Accepted Answers:
Addition of redundant vias

7) Why do we need to make the metal density uniform over the layout by inserting dummy metal fills? *1 point*

- To improve the timing of the circuit
- To reduce switching power dissipation in the circuit
- To tackle signal integrity issues in the circuit
- To improve the manufacturability of the circuit

No, the answer is incorrect.
Score: 0
Accepted Answers:
To improve the manufacturability of the circuit

8) LVS compares the following: *1 point*

- Layout and Schematic (Starting Netlist)
- Inductance and Capacitances
- Logic, Voltage, and Slack
- Logic and Slack

No, the answer is incorrect.
Score: 0
Accepted Answers:
Layout and Schematic (Starting Netlist)

9) Which of the following task is **NOT** a signoff verification task? *1 point*

- Behavior Synthesis
- Static Timing Analysis
- Signal Integrity Checks
- Power Integrity Checks

No, the answer is incorrect.
Score: 0
Accepted Answers:
Behavior Synthesis

10) For IC design, what do we mean by "tape out"? *1 point*

- Measuring the size of an IC using tape
- Taking backup of netlist on a magnetic tape
- Handing over the final IC layout to the foundry for fabrication
- Adding magnetic tape to the IC design

No, the answer is incorrect.
Score: 0
Accepted Answers:
Handing over the final IC layout to the foundry for fabrication